DVOL-Adaptive: A Volatility-Sentiment Driven Bitcoin Trading Strategy

Independent Research Project | Shibabrata Das

1. Introduction

This report details the systematic development, simulation, and optimization of a Bitcoin (BTC) trading strategy over a historical period of approximately 1,000 days. The primary objective was to design a strategy that leverages the Deribit Volatility Index (DVOL) as a sentiment indicator, combined with dynamic risk management parameters, to achieve consistent profitability. The report covers data acquisition, sentiment definition, iterative strategy simulations, parameter optimization (leverage, stop-loss, profit cap), the introduction of dynamic leverage based on extreme volatility, and detailed intraday analysis.

2. Data Acquisition and Preprocessing

2.1 Data Sources

- **Deribit Volatility Index (DVOL):** Fetched daily DVOL data for Bitcoin for the last 1,000 days. This index serves as the primary sentiment driver.
- **Yahoo Finance (BTC-USD):** Acquired daily and hourly Open, High, Low, Close (OHLC) price data for BTC-USD for the corresponding period (up to 3 years daily, 730 days hourly due to API limits).

2.2 Data Merging and Initial Features

The DVOL and BTC-USD datasets were merged on their respective date fields. Key features were defined:

- **Bought_Price:** The opening price of BTC for the day, serving as the entry price for a trade.
- **Price:** The closing price of BTC for the day, used for PnL calculation if no intraday triggers were hit.
- **Sentiment:** Derived from the DVOL index — if today's DVOL was higher than yesterday's, sentiment was classified as 'Bearish'; otherwise 'Bullish'. This assumes rising DVOL implies increasing fear/uncertainty.
- **Verdict:** An initial evaluation of whether the Sentiment prediction aligned with the daily price movement.

All processed data was saved to *BTC_Tracker_1000d.xlsx*.

3. Initial Strategy Simulation (Fixed Parameters)

An initial trading simulation was conducted using fixed parameters:

- Daily Investment: $1,000
- Leverage: 5x
- Stop-Loss (SL): 10%
- Profit Cap (TP): 15%

Trades were executed daily based on Sentiment (long for 'Bullish', short for 'Bearish'). PnL for each day considered whether intraday high/low fluctuations would trigger the stop-loss or profit cap; if neither was hit, PnL was based on Open-to-Close price movement.

Metric	Result

Total Net Profit (after 1,000 days)	~$6,736.52
Average Daily Profit (profitable days)	~$85.00
Average Daily Loss (losing days)	~-$70.60

This simulation served as a baseline for further optimization.

4. Strategy Refinement and Parameter Optimization

4.1 DVOL-based Leverage Advisor

A dynamic leverage mechanism was introduced based on DVOL. The 90th percentile of historical DVOL values (~61.23) was used to identify periods of 'Extreme Volatility'. On days above this threshold, leverage was advised to be reduced to 50% of base leverage; otherwise full leverage was recommended. This advisor was exported to *BTC_DVOL_Leverage_Advisor.xlsx*.

4.2 Single-Parameter Optimization

- **Profit Cap Adjustment:** Increasing the profit cap from 15% to 20% (5x leverage, 10% SL) raised total net profit to ~$9,099.80.
- **Leverage Optimization:** A sweep across 1x–100x leverage (10% SL, 20% PC) revealed an optimal leverage of **7x**, yielding a maximum total net profit of **$9,631.86**. Beyond this, increased leverage led to diminishing returns and eventual significant losses.
- **Stop-Loss Optimization:** With optimal 7x leverage and no profit cap, varying stop-loss from 0.1% to 100% showed that a range of **14%–30%** yielded the best overall profitability — tight stop-losses were often detrimental in this volatile market.

4.3 Advanced Intraday Analysis (Hourly Data Integration)

A key limitation of daily OHLC data is the inability to determine the sequence of intraday events. Hourly BTC-USD data (1-hour interval, 730 days) was fetched and processed to resolve this:

- **Intraday Fluctuation Analysis:** Each day's hourly data was analyzed for the highest/lowest points relative to the day's open, with timestamps, to determine which trigger (stop-loss or profit cap) hit first.
- **Hourly Volatility Matrix:** A comprehensive matrix (*BTC_Hourly_Volatility_Matrix.xlsx*) pivoted hourly high/low % fluctuations for each day, enabling more granular and accurate intraday trade resolution.

5. Extensive Grid Search for Optimal Fixed SL/TP with Dynamic Leverage

Building on the intraday analysis, a grid search was performed to find the optimal combination of base leverage, stop-loss, and profit cap, incorporating the dynamic leverage mechanism.

Grid Search Parameters

- Base Leverage: 1x to 15x (step 1x)
- Stop-Loss %: 1% to 15% (step 1%)
- Profit Cap %: 1% to 20% (step 1%)
- Dynamic Leverage: base leverage halved on days when DVOL exceeds the 90th percentile
- Intraday Resolution: hourly volatility matrix used to precisely simulate SL/TP triggers within each day

Optimal Configuration Identified

Maximum Total Net Profit: $11,816.92

- Base Leverage: 10x (dynamically halved to 5x on extreme volatility days)
- Stop-Loss Percentage: 14.0%
- Profit Cap Percentage: 20.0%

This result demonstrates a significant improvement over previous fixed-parameter simulations, highlighting the value of both intraday resolution and dynamic risk management.

6. Comparison: Fixed Leverage vs. Dynamic Leverage

To quantify the benefit of dynamic leverage, the strategy was re-simulated using the optimal SL/TP parameters (10x leverage, 14% SL, 20% PC) under two scenarios:

Scenario	Total Net Profit	Winning Days
Fixed Leverage (10x)	$9,064.84	50.80%
Dynamic Leverage (10x base, halved on extreme DVOL)	$11,816.92	50.80%

Conclusion: The dynamic leverage approach — reducing exposure during periods of high DVOL — significantly outperformed the fixed leverage strategy, increasing total net profit by approximately $2,752.08 while maintaining the same winning day percentage. This suggests that while the directional Sentiment signal may be accurate, managing risk by reducing position size during high volatility is crucial for overall profitability.

7. Sentiment Confidence and Prediction Accuracy

7.1 Historical Win Rate of DVOL-based Sentiment

- Historical Win Rate: **53.20%**
- Confidence Rating: MODERATE (slight edge over random chance)

7.2 Rolling Win Rate Trend

Plotting the 100-day rolling win rate showed fluctuations over time. The most recent 100-day trend indicated a win rate of **64.00%**, suggesting the sentiment signal may be more effective in recent periods, or that market conditions have favored its accuracy.

7.3 Detailed Prediction Accuracy Analysis (Dynamic Leverage Strategy)

- **Lost Days, Correct Prediction:** 50 days where the sentiment prediction was correct, yet the trade still resulted in a loss — due to adverse intraday movement or stop-loss activation despite the daily direction aligning with sentiment.
- **Winning Days, Wrong Prediction:** 26 days where the sentiment prediction was incorrect, but the trade still yielded a profit — likely due to a quickly-hit profit cap or the dynamic leverage adjustment mitigating losses.

These findings emphasize that while sentiment provides an initial directional bias, the robust risk management framework plays a significant role in determining the final trade outcome.

8. Advanced Strategy with Trailing Stops (Underperformance)

An attempt was made to implement a more advanced strategy incorporating dynamic leverage and an intraday trailing stop-loss (moving to breakeven if price moves 10% in the trade's favor).

Strategy Parameters

- Daily Investment: $1,000
- Base Leverage: 5x (reduced to 2.5x on extreme volatility days)
- Initial Stop-Loss: 10%
- Profit Cap: 20%
- Trailing Stop Activation: 10% favorable move shifts SL to breakeven

Results

- Total Net Profit: **-$302.97**
- Max Drawdown: -26,637.61% (of peak capital) / -$4,618.38 absolute
- Volatility of Daily PnL: $87.60
- Approximate Sharpe Ratio: -0.00

Conclusion: Despite its theoretical appeal, this trailing stop-loss implementation resulted in a net loss over the simulated period, struggling to capture upside while remaining susceptible to losses — suggesting the chosen parameters or mechanics were not optimal for the given market conditions and signal.

9. Conclusion and Future Work

This research successfully demonstrated the development and optimization of a Bitcoin trading strategy using DVOL-based sentiment and dynamic risk management. The most effective strategy identified through extensive grid search — incorporating dynamic leverage and precise intraday SL/TP resolution — yielded a **total net profit of $11,816.92** over the simulated period.

Key Takeaways

- **Dynamic Leverage is Crucial:** Adjusting leverage based on DVOL (reducing during extreme volatility) significantly enhances profitability.
- **Intraday Resolution Matters:** Using hourly data to accurately determine SL/TP triggers is vital for realistic PnL calculation and optimization.
- **Sentiment as a Bias:** DVOL-based sentiment provides a moderate edge, amplified by robust risk management.
- **Careful Parameter Tuning:** Each parameter (leverage, SL, TP) has a substantial impact on profitability and requires thorough optimization.

Future Work

- Refine trailing stop logic — investigate alternative implementations and parameters.
- Enhance the sentiment signal — explore other volatility indicators or machine learning-based prediction.
- Incorporate realistic transaction costs and slippage into simulations.
- Extend the strategy to a portfolio of cryptocurrencies or assets.
- Robustness testing across different market regimes and out-of-sample data.